MCT232 [Electronics for Instrumentation]

Major Task (Cooling Pad)

Team 21

Name	ID
Ashraf Ahmed	2200403
Adham Sherif	2200445
Nassir Afifi	2200697

Submitted to: Dr. Maged Ghoneima

Introduction :

The Cooling Pad Project aimed to design and develop an efficient, portable cooling pad to enhance user comfort for laptops by dissipating heat effectively. The primary goal was to create a lightweight, energy-efficient device with adjustable cooling levels, targeting students, professionals, and gamers who rely on laptops for extended periods. The project involved ideation, prototyping, testing, and refinement, integrating advanced materials and fan technology to optimize thermal performance while maintaining affordability and aesthetic appeal.

Why we need cooling pad ?

A cooling pad helps dissipate heat from laptops and electronic devices ,It prevents overheating, which can cause performance issues or hardware damage, By improving airflow, it keeps internal components like the CPU and GPU cooler ,Lower temperatures help avoid thermal throttling and ensure stable performance Using a cooling pad can extend the lifespan of your device ,Many pads also improve ergonomics by raising the laptop for better airflow and posture.

Research :

→ We choose the AM2320 digital sensor as a Tempreture sensor because

1)it has a long-term stability ,

2)long distance signal transmission ,

3)precise calibration ,

4)low power consumption ,

5)communication free choice

→We control the speed of the fan through a circuit that consists of transistor , diode and 1K ohm resistance

→We use the first potentiometer as we can adjust the brightness of the screen

→The second potentiometer used to switch between Automatic and Manual mode

→The LED needed to make an alert when tempreture exceed the safe level

→Also , We need a switch to turn off , on the fan as we need

→The LCD screen needed to display the tempreture degree and speed of fan (size 16x2 or higher)

→We will use a 5V fan because the control of speed and their implementation is simple

Final Components used in project :

1) Lcd screen 16x4
2) Tempreture sensor AM2320
3) 2x Potentiometer 10k ohm
4) Fan 5v
5) Arduino Mega
6) LED Diode
7) Diode 1N4007
8) Transistor NPN (2N2222A)
9) 2x Resistors 1K
10) Button

Concept(How system can operate) :

- Sensor reads temperature.
- the AM2320 returns a digital value directly in degrees Celsius, no voltage conversion or math needed
- The signal sent to microcontroller
- Microcontroller processes this data.
- Based on a set threshold, it turns the fan on (according to the tempreture , Tempreture $\uparrow$, Speed $\uparrow$)
- If tempreture exceeds the limit tempreture , the led turns on
- Real-time display of the temperature and the speed of the fan

- **The switch determines if :**

When OFF→ the fan is stopped and temperature is still displayed.

When ON→the system proceeds to read and adjust the fan.

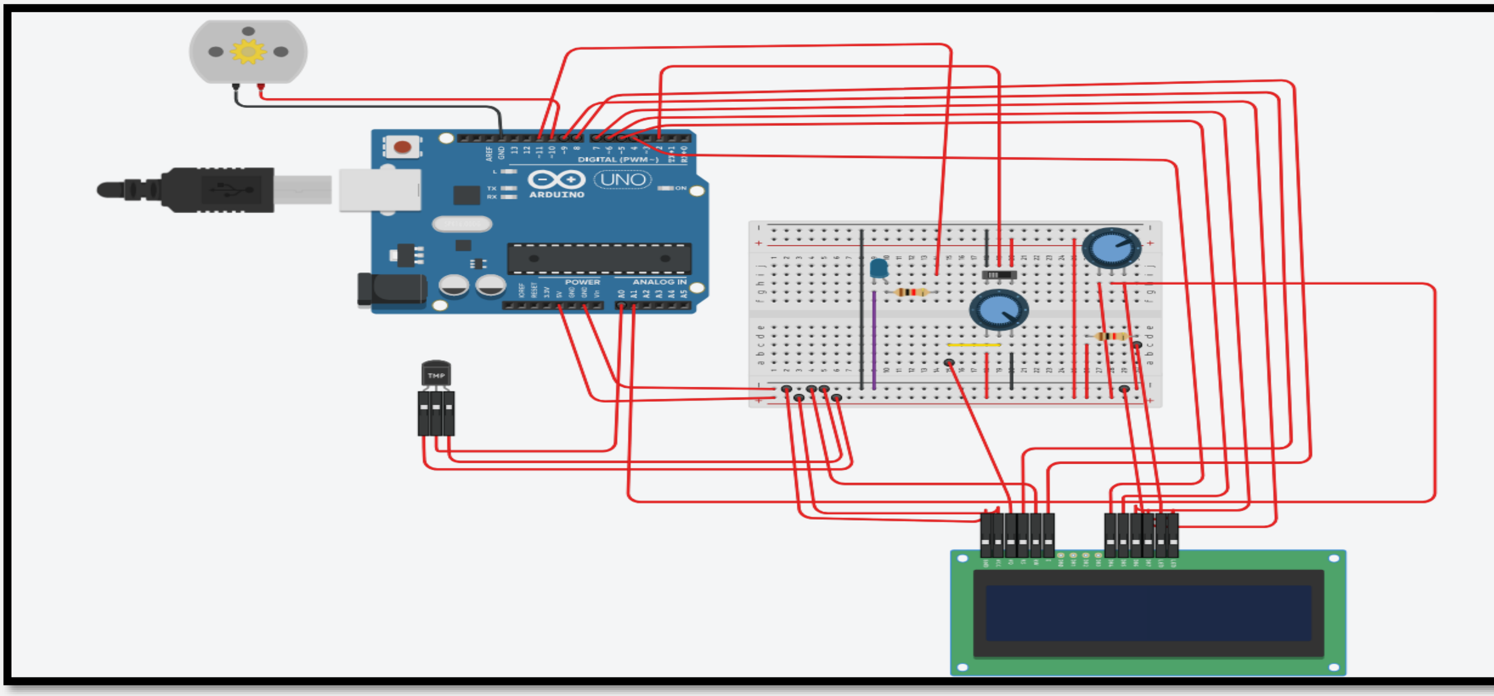
- **If the potentiometer reading is > 512, it's Auto Mode. Otherwise, it's Manual Mode.**

- **The LCD shows four rows:**

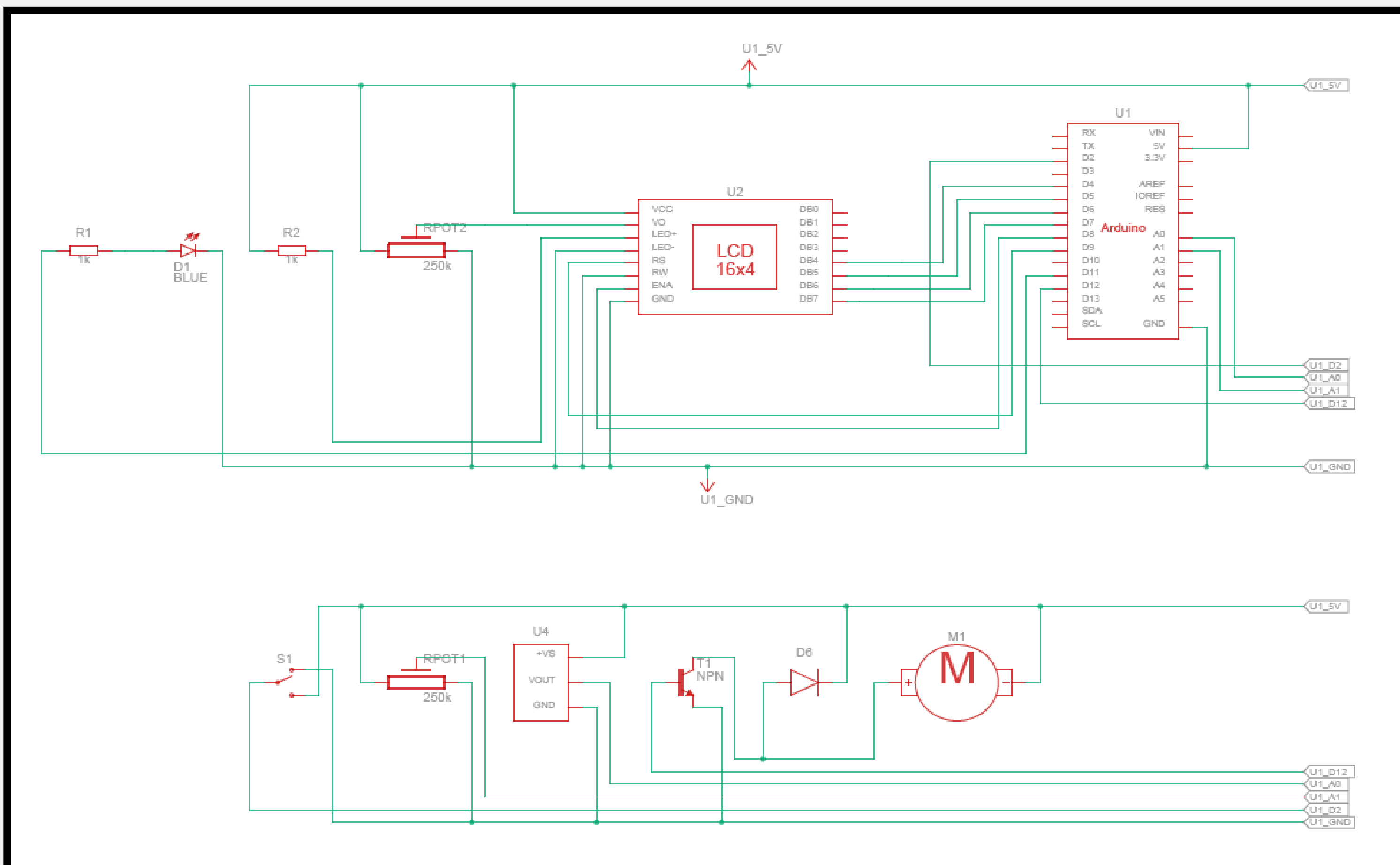
  Row 1: System mode and temperature.
  Row 2: Fan speed
  Row 3: Humidity percentage
  Row 4:Message that show system status
  (If the fan is off , it still displays the temperature.)

- **If temperature exceeds 27°C, the LED turns on.**

Circuit Diagram :



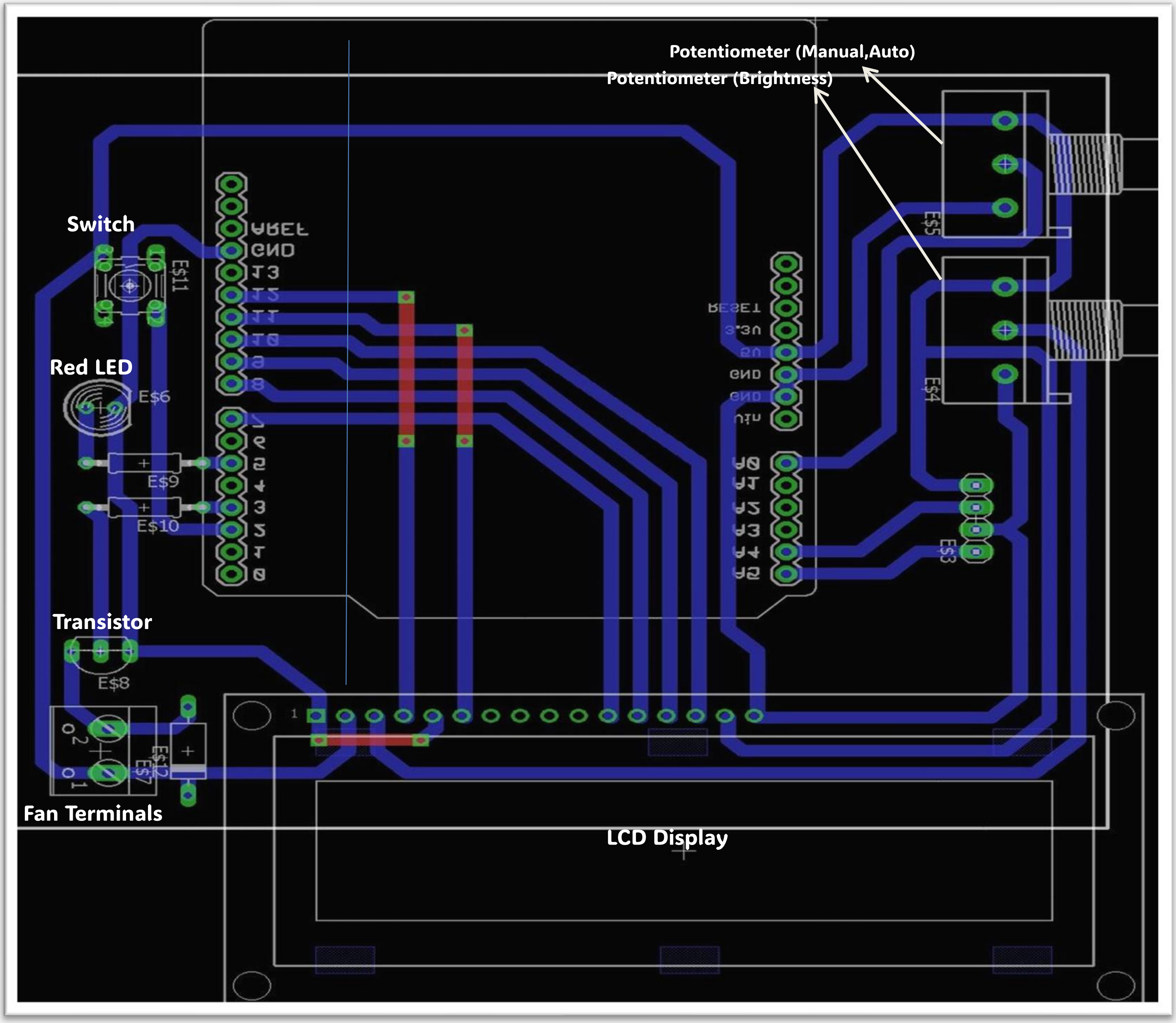
Circuit implementation :



PCB Design :

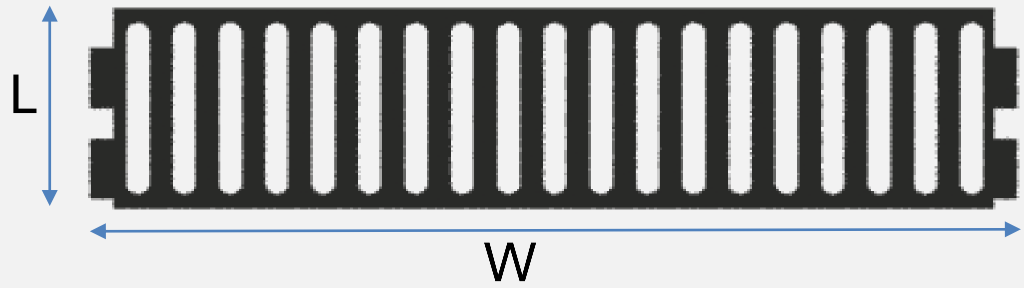
Making a PCB improves reliability, reduces wiring mess, and ensures compact, organized circuits. It enables easier replication for production and enhances electrical performance. Also, it offer better safety and durability compared to breadboards. They also give our project a professional appearance and help build valuable design skills.

Circuit implementation by PCB :



Casing Design :

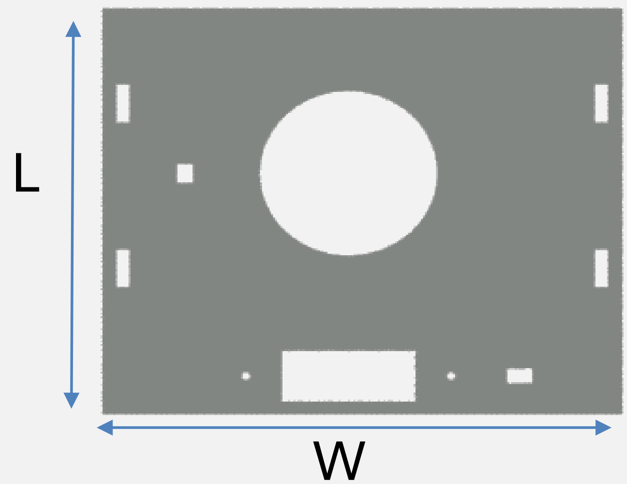
1) Back Plate



Length = 100 mm

Width = 360 mm

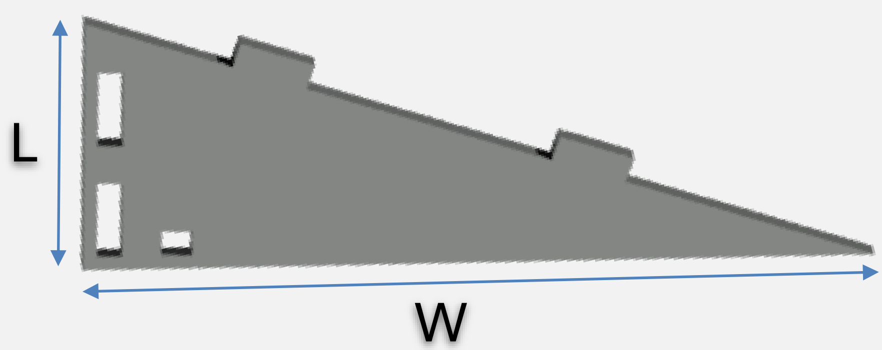
2) Top Plate



Length = 320 mm

Width = 320mm

3) (2x) Side Plate



Length = 100 mm

Width = 305 mm

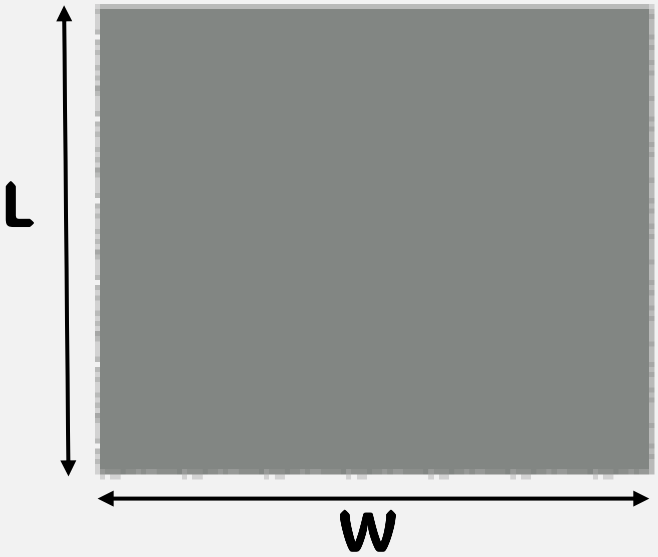
4) Fan Cover

Diameter = 125 mm

5) Fan Base

Inner Diameter = 50

6) Bottom

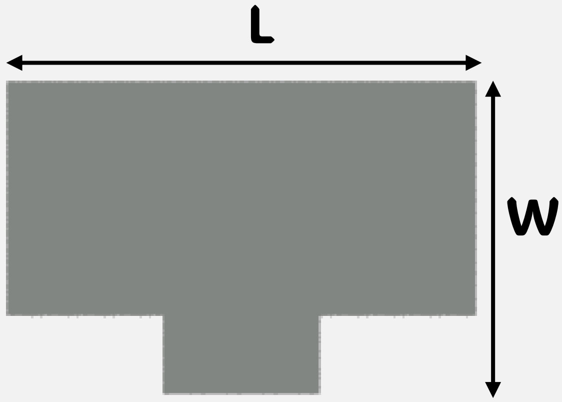


Length = 360 mm

Width = 305 mm

7) Fan base support

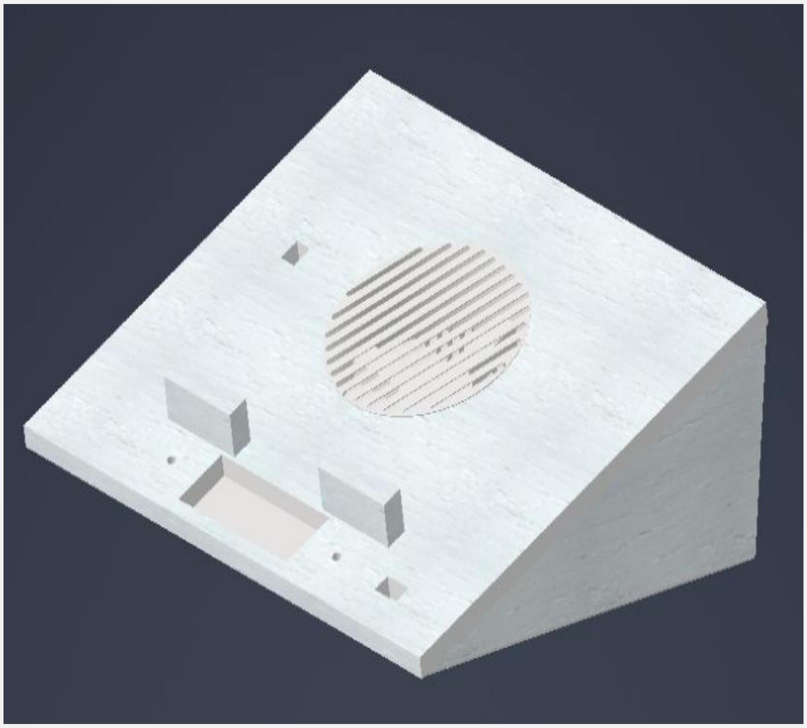
8) Lap Support

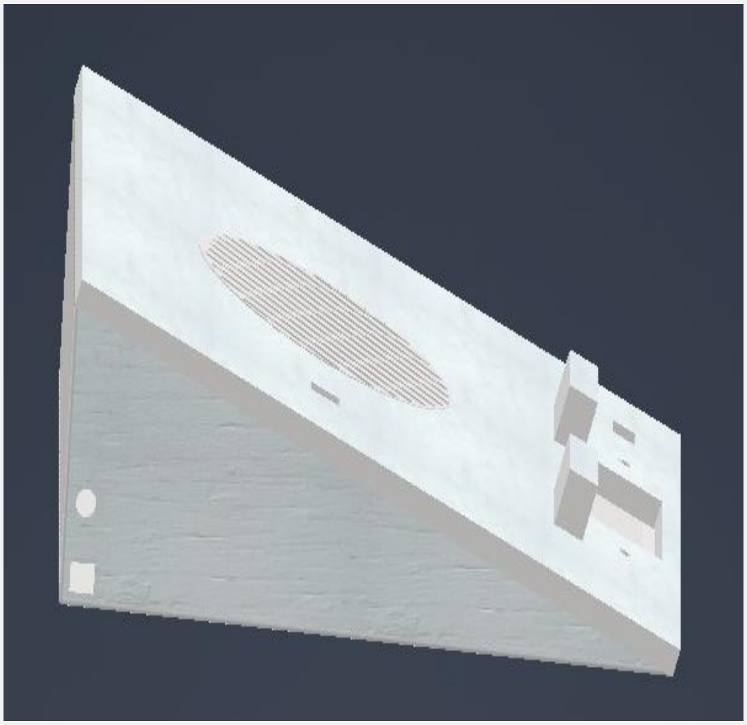


Length = 60 mm

Width = 40 mm

Final Product :





Testing the Cooling Pad :

All tests done under Reservoir Temperature = 25 C

May be some errors due to delays in the code

Cooling Rate = $\left(\frac{Diffrnece\ in\ tempreture}{Time} \right)$ **C/sec**

First: Test cooling from 40 C to 35 C

Time = 72 sec

Cooling rate = 0.07 C/sec

Second: Test cooling from 35 C to 30 C

Time = 70 sec

Cooling rate = 0.071 C/sec

Third: Test cooling from 30 to 25 C

Time = 62 sec

Cooling rate = 0.0806 C/sec

Average Cooling Rate = (0.2216) / 3 = 0.074 C/sec

= 4.44 C/min

Final Hardware Set-up :

→Comparison of the fan project with other market fans

Feature	Noctua F12	Fan Project	Noctua A4x10	DC Fan 0.12A (Micro Ohm)
Size	120×120×25	120×120×25	40×40×10	50×50×12
Airflow (CFM)	55	45	4.83	10
Noise Level (dB)	22.4	25	17.9	28
Power Consumption	0.75 W	1.2 W	0.25 W	1 W
USB Compatibility	Yes	Yes	NO	NO
Price	13.95 $	4.95 $	4.98 $	2.98 $

→Compare full project with other cooling pads in market

Comparison	Project	Redragon GCP500 IVY	Crush C1000	Tecno I-10
DESIGN				
Average cost(EGP)	550	600	500	750
Features	1) lcd display 2)led alert 3) full CTRL on fan 4) temperature sensor	1) high cooling rate 2) RGB lights 3) 4)	1) RGB lights 2) inclined stand 3) light weight 4)	1) high cooling rate 2) stand with high loads 3)
Limitations	1) low cooling rate 2) not fit all sizes 3) not stand with high loads	1)incompatibility 2) high noise 3) high power consumption	1) low cooling rate 2) Dust Accumulation 3)weak body	1) Dust Accumulation 2) high noise 3) low efficiency

Features :

→ Adapts to changing temperatures in Auto Mode.

→ Gives user full control in Manual Mode.

→ Alerts users to unsafe heat conditions.

→ Display the system state on the LCD screen.

→ Switch to on/off the Fan

→ USB Power Port

Limitations of the Cooling Pad :

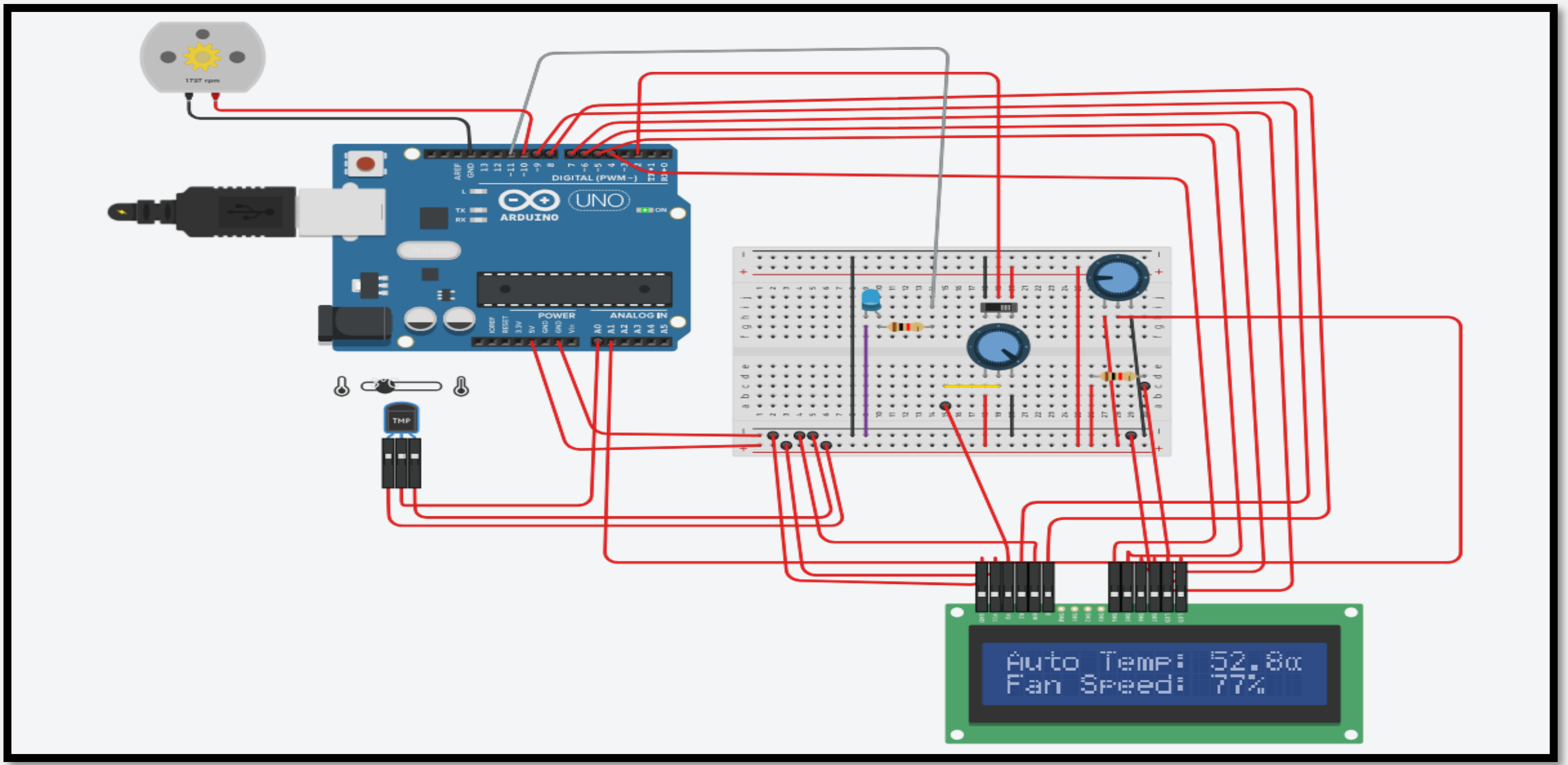
1) Limited Cooling Effectiveness
2) Incompatibility (Not all pads fit every laptop size)
3) Dust Accumulation (They can draw in dust, which may close laptop vents over time)

How can we upgrade the Cooling Pad ?

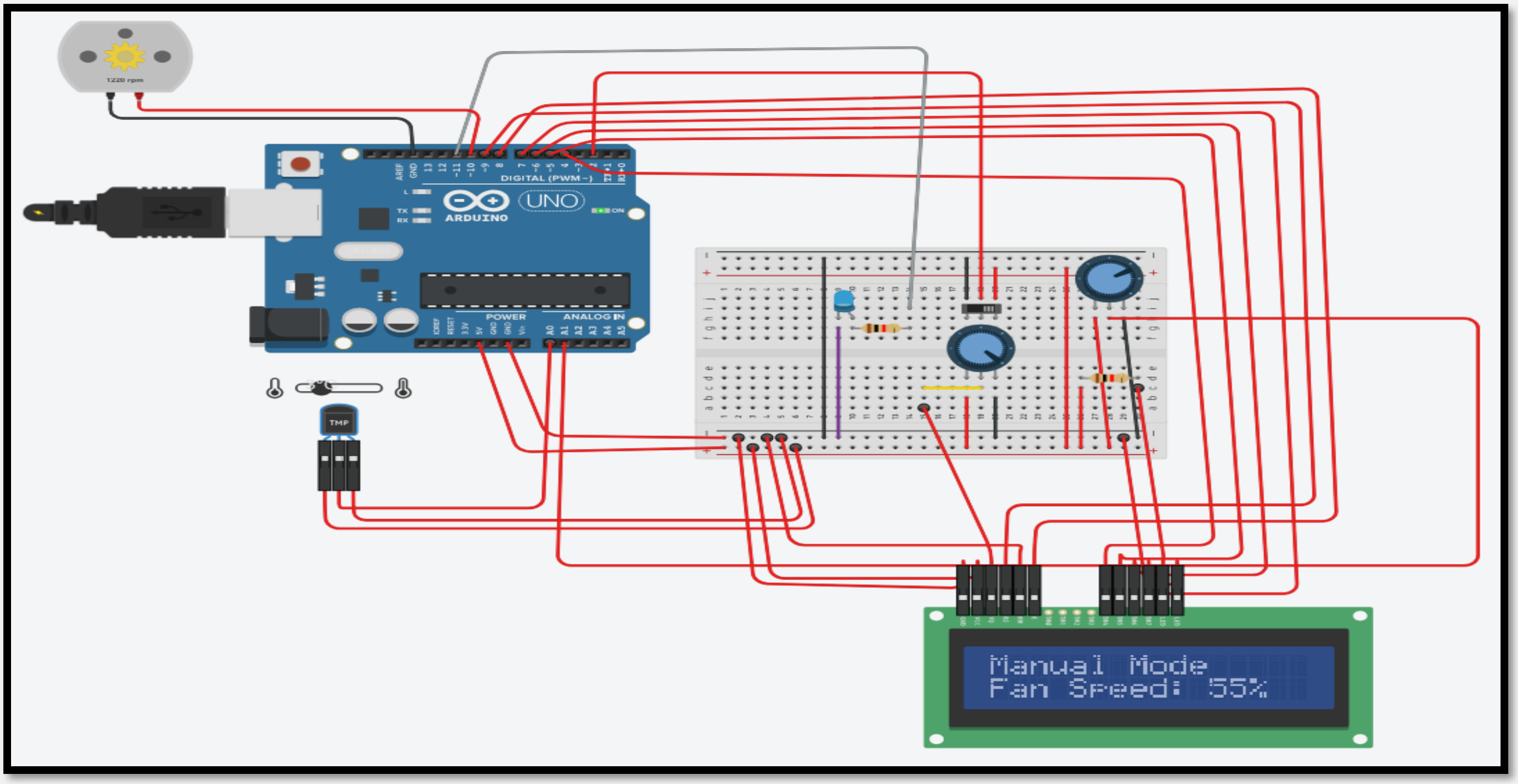
1) Enhanced Material (Use carbon fiber or graphene-based composites for a lighter, stronger)
2) Upgrade to brushless DC fans with advanced blade designs to minimize noise
3) Install built-in rechargeable battery to reduce reliance on external power sources

Software Simulation :

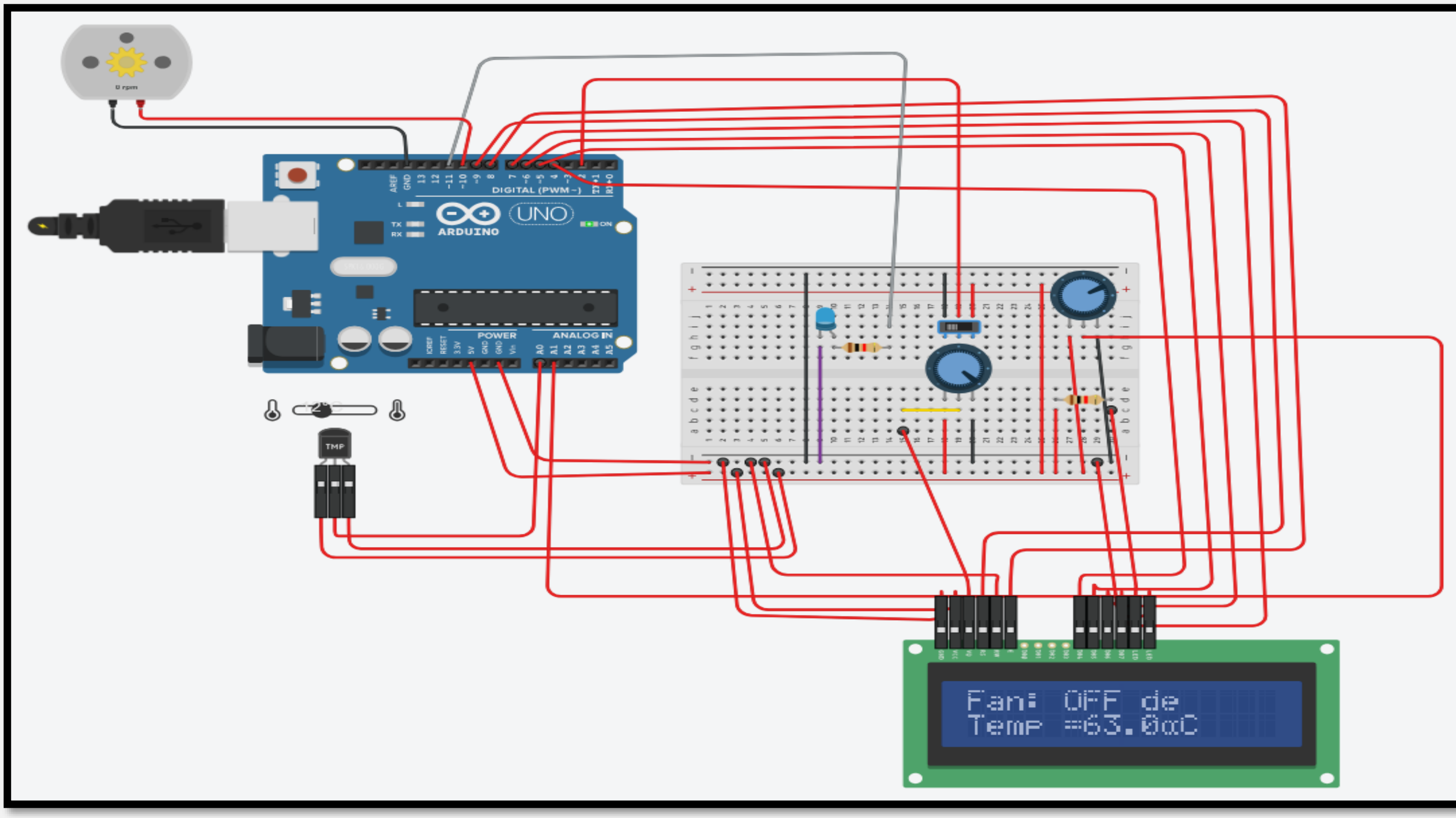
First : Automatic Mode



Second : Manual Mode



Third: Fan switch is off



Conclusion :

The Cooling Pad Project successfully delivered a functional, user-friendly product that met its core objectives of enhancing laptop performance and user comfort. Despite initial challenges, the iterative process, informed by user feedback and rigorous testing, resulted in a reliable and market-competitive cooling pad. The lessons learned—particularly the value of material optimization, cross-functional collaboration, and agile project management—provide a strong foundation for future innovations. This project not only achieved its technical goals but also reinforced the importance of adaptability and user-centric design in product development.

Sources of project :

Arduino Code

https://drive.google.com/file/d/19-vNaQ0ISgxV1hH8MtZ3cE7LCn0z7euB/view?usp=drive_link

Hardware Simulation Video

https://drive.google.com/file/d/1ehIReWIF7IQdxlrMfv70KrSIUAemyF0U/view?usp=drive_link